Etienne Mueller

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Innovative AI and computational neuroscience professional with a history of developing advanced algorithms, machine learning systems, and neuromorphic computing solutions across academia, startups, and leading global organizations. Proven track record of designing efficient AI models/tools that reduce processing times, enhance accuracy, and lower costs. Skilled in conducting AI research, designing/deploying cutting-edge tools, and using high-powered computing to analyze complex data sets.

Professional Experience

Assistant Professor AI & Data Science (*Carnegie Mellon | KMITL (CMKL), Thailand*) **2025 – Present**

- Investigating hierarchical spiking neural networks for cognitive neuromorphic computing, focusing on bio-inspired multi-timescale architectures for compositional reasoning with planned validation on various neuromorphic hardware
- Developing the curriculum for *Neural Networks and Deep Learning*, *LLMs and Agentic AI*, and *Big Data Computing*

Postdoctoral Researcher AI (*University of Melbourne, Australia*) **2023 – 2025**

- Developed neural network growth algorithms to create biologically-inspired memory cells for more efficient recurrent neural networks, implementing ML simulations of brain imaging data across different developmental stages using JAX
- Ran deep convolutional neural networks on a Slurm-based HPC with up to 4xH100 GPUs per node for automated segmentation of synchrotron brain imaging data, reducing the need for manual annotation by a factor of 10

AI Engineer (*Flowers Software, Germany*) **2022 – 2023**

- Developed a TensorFlow-based information extraction workflow on AWS to identify recurring positions on invoices that complies with EU data privacy law, saving the company over €5,000/month by eliminating third-party API costs

AI Researcher (*Infineon Technologies, Germany*) **2018 – 2022**

- Developed a TensorFlow-based toolbox for converting conventional to spiking neural networks, which was subsequently used in a research project to reduce the simulation time of hardware components in neuromorphic systems by half

Co-founder & CEO (*Slive Technologies, Germany*) **2015 – 2016**

- Developed smart wearable devices and location-based algorithms for hands-free data use in industrial environment
- Secured the Nissen Foundation Start-Up Grant (€3,000) to support early-stage product development and business growth

Education

Ph.D. in Computer Science (Artificial Intelligence) (*Technical University Munich, Germany*) **2023**
M.Sc. in Product Development (Autonomous Driving) (*Technical University Hamburg, Germany*) **2017**
B.Sc. in Mechanical Engineering (Robotics) (*Technical University Hamburg, Germany*) **2014**

Skills

Programming Python, TensorFlow, JAX, Bash, Swift, Xcode, C++, SQL
Languages German (native), French (native), English (fluent), Spanish (basic), Chinese (basic)

Open Source Projects

High-Performance Zebrafish (HPZ)

- End-to-end pipeline that consolidates multiple manual steps for loading, preprocessing, and detecting neurons and spikes in microscopy data into a single automated process, reducing manual intervention and error by a factor of five

Convert2SNN

- A TensorFlow-based library that converts conventionally trained neural networks with continuous activation functions to spiking neural networks, with minimal to no performance loss, to estimate energy consumption in neuromorphic systems

Selected Publications

E. Mueller, W. Qin, "Reverse Engineering Neural Connectivity: Mapping Neural Activity Data to Artificial Neural Networks for Synaptic Strength Analysis," in *8th International Conference on Information Technology (InCIT),* Chonburi, Thailand and Kanazawa, Japan, 2024, pp. 451-454.

E. Mueller, V. Studenyak, D. Auge, A. Knoll, "Spiking Transformer Networks: A Rate Coded Approach for Processing Sequential Data," in *7th Int. Conference on Systems and Informatics (ICSAI),* Online (Jiaxing, China), 2021, pp. 1-5.

See Google Scholar profile for a full list.